Topic 6

Frequency Response of FIR Filters

- Sinusoidal response of FIR systems
- Magnitude and phase response
- Delay, first difference, cascades, moving average
- Filtering sampled continuous-time signals

$$H(e^{j\hat{\omega}})$$

Key idea. Frequency response tells how an FIR filter changes each sinusoidal component.

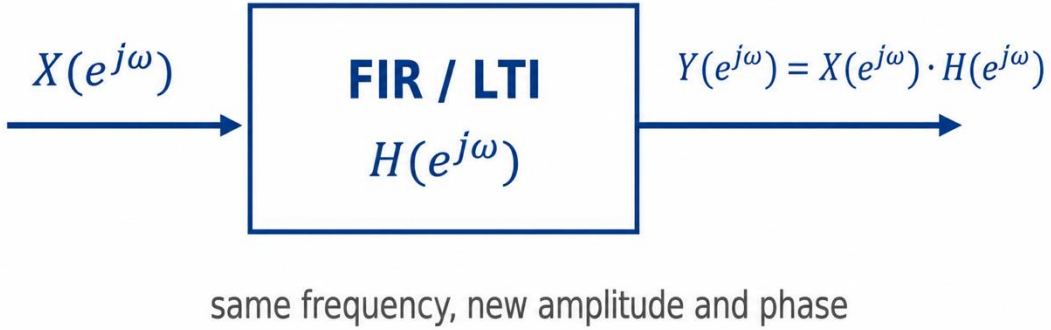
Sinusoidal response of FIR systems

An LTI filter preserves the frequency of a complex exponential

$$y[n] = \sum_{k=0}^{M} b[k]\, x[n-k]$$

$$x[n] = X\,(e^{j\hat{\omega}n})$$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{M} b[k]\, e^{-j\hat{\omega}k}$$



- The frequency does not change.
- The complex gain H changes amplitude and phase.

Key idea. For LTI systems, sinusoids pass through as sinusoids at the same frequency.

Example 1: two-point moving average

Compute the frequency response, magnitude and phase from an FIR filter

$$y[n] = \tfrac{1}{2}x[n] + \tfrac{1}{2}x[n-1] \qquad h[n] = [\tfrac{1}{2}, \tfrac{1}{2}]$$

1 **Identify the filter coefficients**

$h[0] = \tfrac{1}{2}, \quad h[1] = \tfrac{1}{2}$ (zero elsewhere)

2 **Apply a complex sinusoid**

$x[n] = e^{j\hat{\omega}n}$

$y[n] = e^{j\hat{\omega}n}\left(\tfrac{1}{2} + \tfrac{1}{2}e^{-j\hat{\omega}}\right)$

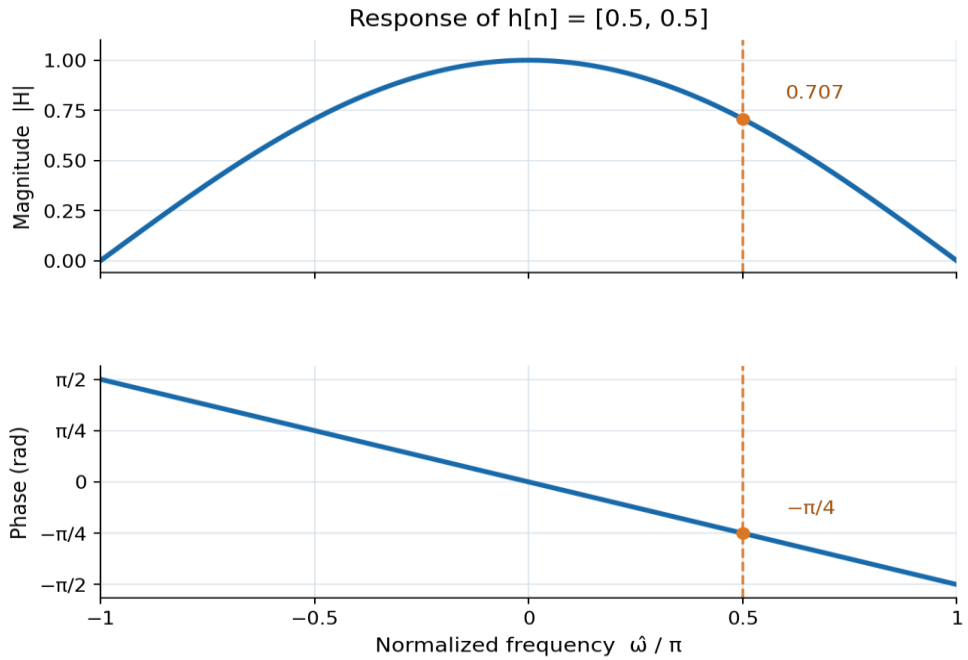
3 **Read off the complex frequency response**

$H(e^{j\hat{\omega}}) = \tfrac{1}{2}(1 + e^{-j\hat{\omega}})$

4 **Factor and obtain magnitude and phase**

$H(e^{j\hat{\omega}}) = e^{-j\hat{\omega}/2}\cos(\hat{\omega}/2)$

$|H| = \cos(\hat{\omega}/2), \quad \angle H = -\hat{\omega}/2$



At $\hat{\omega} = \pi/2$: $|H| = 0.707$, $\angle H = -\pi/4$ rad (for $-\pi < \hat{\omega} < \pi$; phase undefined at $\pm\pi$)

Key idea. A delay adds a frequency-dependent phase factor; the weighted sum gives H.

From $H(e^{j\hat{\omega}})$ to real sinusoidal output

Magnitude scales; phase shifts

$$H(e^{j\hat{\omega}}) = |H(e^{j\hat{\omega}})|\, e^{j\angle H(e^{j\hat{\omega}})}$$

$$A\cos(\hat{\omega}n+\varphi) \rightarrow A|H(e^{j\hat{\omega}})|\cos(\hat{\omega}n+\varphi+\angle H(e^{j\hat{\omega}}))$$

- The output amplitude is multiplied by the magnitude response.
- The output phase is shifted by the phase response.
- A zero in H cancels that sinusoidal component.

Key idea. The frequency response is a lookup table for every sinusoidal component.

Example 2: sinusoidal output

Use the frequency response calculated on the previous slide

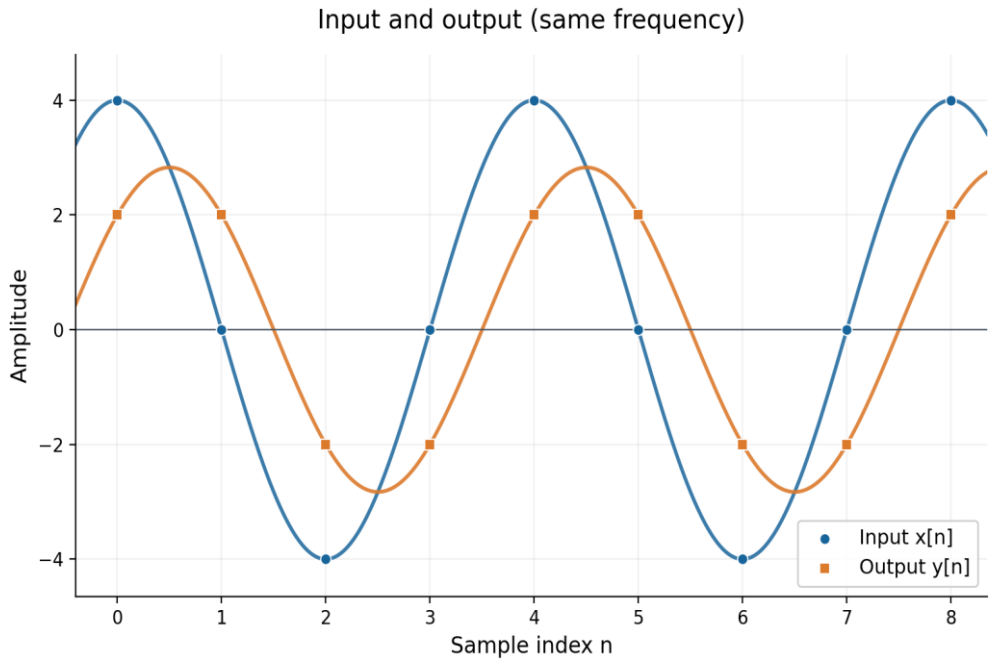
$$x[n] = 4\cos(\pi n/2) \qquad H(e^{j\hat{\omega}}) = \tfrac{1}{2}(1 + e^{-j\hat{\omega}})$$

1. Identify the input sinusoid
   $A = 4, \quad \hat{\omega}_0 = \pi/2, \quad \varphi = 0$

2. Evaluate the given frequency response
   $H(e^{j\pi/2}) = \tfrac{1}{2}(1 - j) = 0.707\, e^{-j\pi/4}$

3. Read the gain and phase shift
   Gain $= 1/\sqrt{2} \approx 0.707$; phase $= -\pi/4$ rad

4. Compute the output parameters
   $A_{out} = 4/\sqrt{2} = 2\sqrt{2}, \quad \varphi_{out} = -\pi/4$



$$\mathbf{y[n] = 2\sqrt{2}\cos(\pi n/2 - \pi/4) \approx 2.828\cos(\pi n/2 - 0.785)}$$

Key idea. Given $H(e^{j\hat{\omega}})$, evaluate it at the input frequency, then scale the amplitude and add the phase shift.

Superposition and frequency response

A sum of sinusoids is filtered component by component

$$x[n] = \sum_{r} X_r \, e^{j\hat{\omega}_r n}$$

$$y[n] = \sum_{r} X_r \, H(e^{j\hat{\omega}_r}) \, e^{j\hat{\omega}_r n}$$

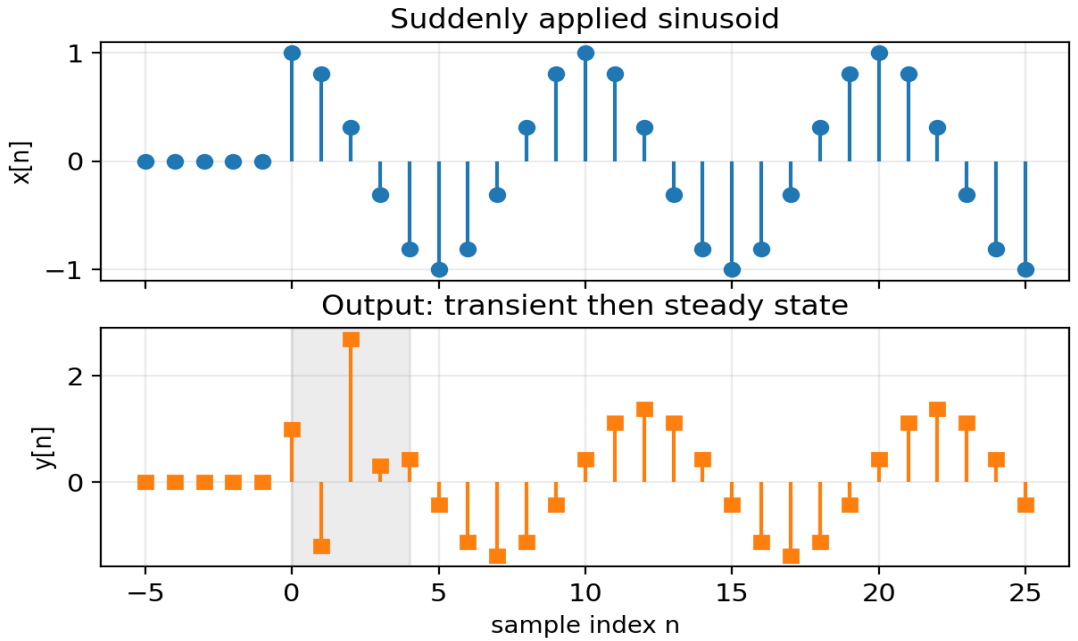
- Linearity lets us filter each component separately.
- No new frequencies are created by an LTI FIR filter.
- Components can be amplified, delayed, attenuated, or cancelled.

Key idea. Frequency response maps an input spectrum into an output spectrum.

Steady-state and transient response

A sinusoid that starts at a finite time produces a transient

- Before the input begins, the output is zero.
- During the first M samples, the FIR memory is only partly filled.
- After the transient, the sinusoidal steady-state formula applies.
- When the input ends, a second transient occurs.



Key idea. The sinusoidal-response rule describes the steady state, not the onset transient.

Properties of the frequency response

Periodicity and symmetry

$$H(e^{j(\hat{\omega}+2\pi)}) = H(e^{j\hat{\omega}})$$

$$\text{real } h[n] \Rightarrow H(e^{-j\hat{\omega}}) = H^*(e^{j\hat{\omega}})$$

- Discrete-time frequency response is periodic with period 2π.
- For real coefficients, magnitude is even.
- For real coefficients, phase is odd, except for wrapping and zeros.
- Usually one interval, such as $-\pi \le \hat{\omega} \le \pi$, is enough.

Key idea. The whole response is determined by one principal frequency interval.

Delay system

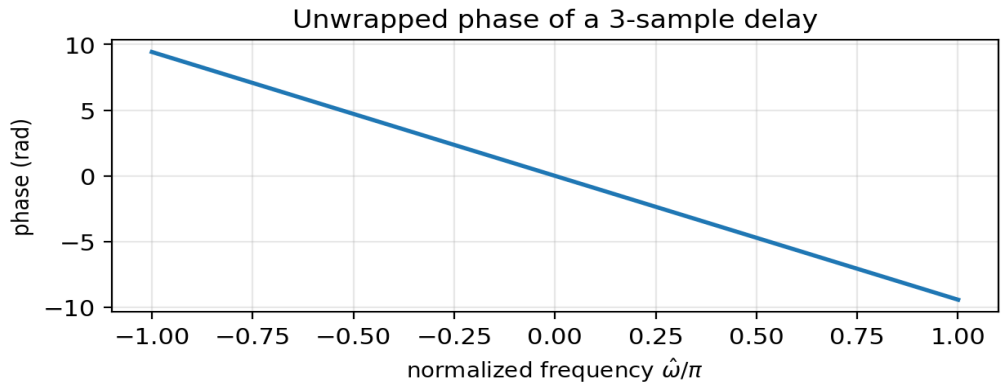
A pure delay changes phase but not magnitude

$$y[n] = x[n - n_0]$$

$$h[n] = \delta[n - n_0]$$

$$H(e^{j\hat{\omega}}) = e^{-j\hat{\omega}n_0}$$

- Magnitude: $|H| = 1$ for all frequencies.
- Phase: $\angle H = -\hat{\omega}n_0$ before wrapping.
- The slope of phase corresponds to delay.



Key idea. A pure delay preserves amplitude while imposing a linear phase shift.

Example 3: pure delay

Compute the output of a delayed sinusoid

$$y[n]=x[n-3], \quad x[n]=2\cos(0.4\pi n+\pi/6)$$

1. $H(e^{j\hat{\omega}}) = e^{-j3\hat{\omega}}$.
2. At $\hat{\omega}=0.4\pi$: phase shift $= -3(0.4\pi)=-1.2\pi$.
3. The amplitude is unchanged: $|H|=1$.
4. $y[n]=2\cos(0.4\pi n+\pi/6-1.2\pi)$.

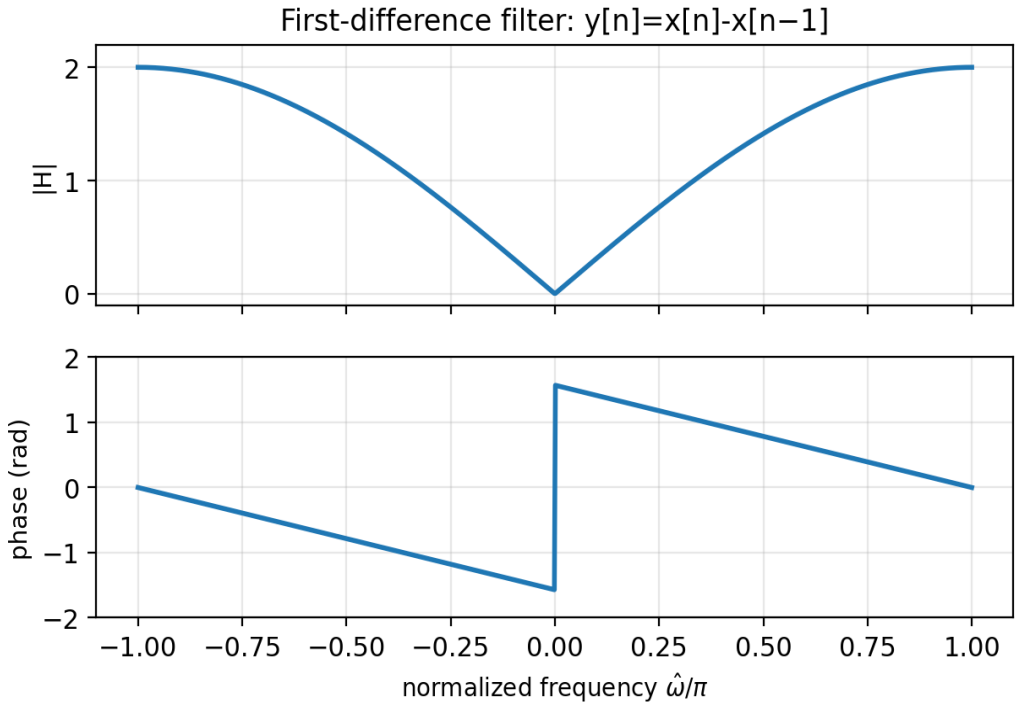
First-difference system

A simple high-pass FIR filter

$$y[n] = x[n] - x[n-1]$$

$$H(e^{j\hat{\omega}}) = 1 - e^{-j\hat{\omega}}$$

$$|H(e^{j\hat{\omega}})| = 2|\sin(\hat{\omega}/2)|$$



Key idea. The first difference removes DC and emphasizes rapid changes.

Example 4: first difference

1. $H(e^{j\hat{\omega}}) = 1 - e^{-j\hat{\omega}} = [1 - \cos(\hat{\omega})] + j\sin(\hat{\omega})$
2. $|H|^2 = [1 - \cos(\hat{\omega})]^2 + \sin^2(\hat{\omega}) = 2 - 2\cos(\hat{\omega})$
3. $|H| = \sqrt{[2 - 2\cos(\hat{\omega})]} = 2|\sin(\hat{\omega}/2)|$

EVALUATE AT THREE FREQUENCIES

$\hat{\omega} = 0$ (DC)

$|H| = 2|\sin(0)| = 0$

Constant signals removed

$\hat{\omega} = \pi/2$

$|H| = 2|\sin(\pi/4)| = \sqrt{2}$

Gain ≈ 1.414

$\hat{\omega} = \pi$

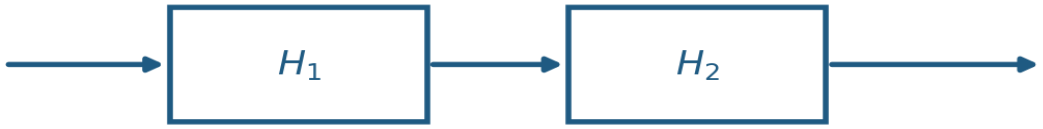
$|H| = 2|\sin(\pi/2)| = 2$

Maximum gain

Key idea. Gain increases from 0 to 2 over $0 \le \hat{\omega} \le \pi \Rightarrow$ high-pass behaviour.

Cascaded LTI systems

Frequency responses multiply; impulse responses convolve



$$H_{eq}(e^{j\hat{\omega}}) = H_1(e^{j\hat{\omega}})H_2(e^{j\hat{\omega}})$$

$$h_{eq}[n] = h_1[n] * h_2[n]$$

- Cascade order does not matter for LTI systems.
- Zeros of either subsystem become zeros of the cascade.
- In time domain, the equivalent impulse response is $h_1 * h_2$.

Key idea. Cascades are easier to analyze in the frequency domain because multiplication replaces convolution.

Example 5: cascade

Two moving sums in sequence

$$h_1[n]=\{1,1\}, \quad h_2[n]=\{1,1\}$$

1. Frequency responses: $H_1=1+e^{-j\hat{\omega}}$, $H_2=1+e^{-j\hat{\omega}}$.
2. Overall response: $Heq=(1+e^{-j\hat{\omega}})^2$.
3. Equivalent impulse response: $heq=h_1*h_2=\{1,2,1\}$.
4. The cascade is a three-point triangular smoother.

Key idea. Cascading two simple averagers produces a stronger smoother.

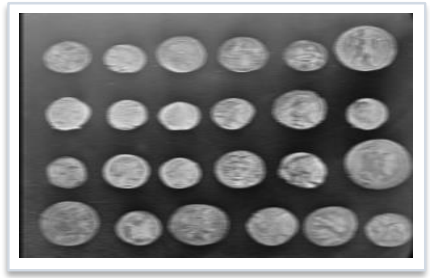
DSPFirst demo: cascade order does not matter

A 5-point average followed by a first difference — or the reverse order



Input image x[n]

5-point average



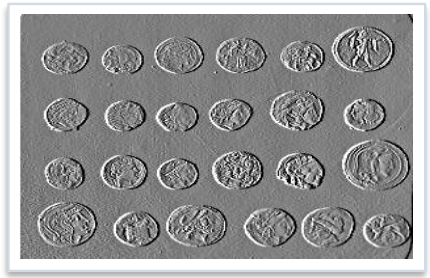
v[n] • smoothed

First difference



$y_1[n]$ • final

First difference



w[n] • differences

5-point average



$y_2[n]$ • final

Explore the original DSPFirst cascade demo ↗

$$H_1H_2 = H_2H_1 \Rightarrow y_1[n] = y_2[n]$$

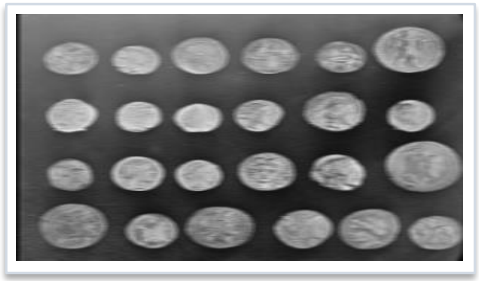
DSPFirst demo: complementary filters

Split an input into a smoothed part and a high-frequency part; then add them



Input image x[n]

H_1



Low-pass output

H_2



High-pass output

+



Reconstructed y[n]

Open DSPFirst demo ↗

$$H_2 = 1 - H_1 \quad \Rightarrow \quad H_1 + H_2 = 1 \quad \Rightarrow \quad y[n] = x[n]$$

High-pass = input − low-pass. Exact reconstruction requires complementary filters and correct alignment.

Moving-average filtering

A low-pass smoother with structured zeros

$$y[n] = (1/L) \sum_{k=0}^{L-1} x[n-k]$$

$$h[n] = 1/L, \quad 0 \leq n \leq L-1$$

- A longer average smooths more strongly.
- The main lobe becomes narrower.
- The phase corresponds to a delay of (L−1)/2 samples.
- Zeros cancel specific sinusoidal components.

Key idea. A moving average is simple in time domain but very informative in frequency domain.

Moving-average frequency response

Step-by-step derivation using a finite geometric series

1 **Specify the L-point moving average**

$$h[k] = 1/L \quad \text{for } 0 \le k \le L-1 \text{ (zero elsewhere)}$$

2 **Insert h[k] into H and expand the finite sum**

$$H(e^{j\hat{\omega}}) = (1/L)\left[1 + e^{-j\hat{\omega}} + \cdots + e^{-j(L-1)\hat{\omega}}\right]$$

3 **Recognize the geometric series with $r = e^{-j\hat{\omega}}$**

$$1 + r + \cdots + r^{L-1} = (1 - r^{L})/(1 - r) \qquad (r \neq 1)$$

4 **Substitute r into the geometric-series formula**

$$H(e^{j\hat{\omega}}) = (1/L) \times (1 - e^{-jL\hat{\omega}}) \,/\, (1 - e^{-j\hat{\omega}})$$

5 **Use the identity for $\theta = L\hat{\omega}$ (numerator) and $\theta = \hat{\omega}$ (denominator)**

$$1 - e^{-j\theta} = 2j\, e^{-j\theta/2} \sin(\theta/2)$$

RESULT · COMPLEX FREQUENCY RESPONSE

$$H(e^{j\hat{\omega}}) = [\sin(L\hat{\omega}/2) \,/\, (L \sin(\hat{\omega}/2))]\, e^{-j\hat{\omega}(L-1)/2}$$

$$|H| = |\sin(L\hat{\omega}/2)| \,/\, [L\,|\sin(\hat{\omega}/2)|] \qquad \text{phase factor} = e^{-j\hat{\omega}(L-1)/2}$$

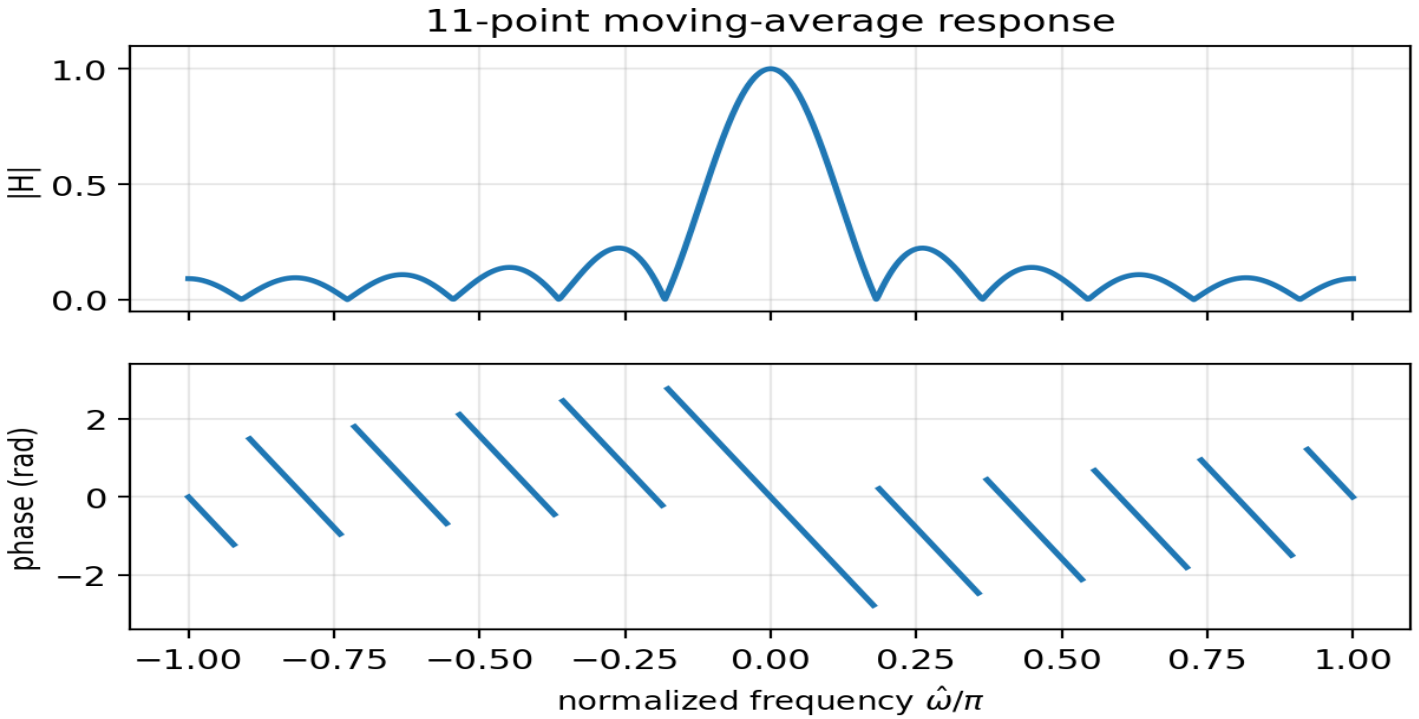
At $\hat{\omega} = 0$, $H = 1$ (by continuity); phase has π jumps where the sine ratio changes sign; undefined at zeros.

11-point moving-average response

Magnitude response shows low-pass smoothing

11-point average: $y[n] = (1/11)\sum_{k=0}^{10} x[n-k] \quad (h[k] = 1/11,\ 0 \le k \le 10)$

$H(e^{j\hat{\omega}}) = [\sin(11\hat{\omega}/2) / (11\sin(\hat{\omega}/2))]\, e^{-j5\hat{\omega}} \quad (H = 1 \text{ at DC})$

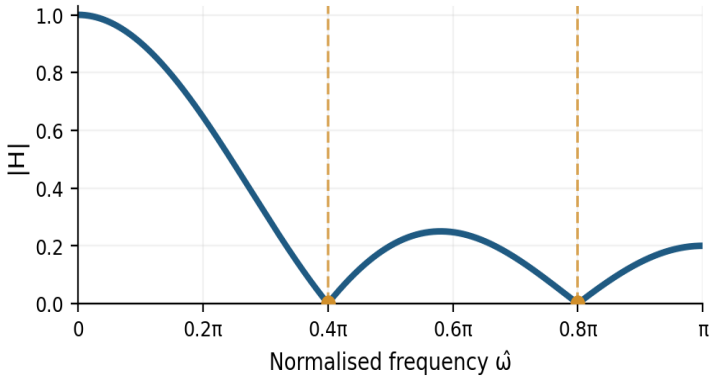


Key idea. Low frequencies near DC pass; many higher frequencies are attenuated or cancelled.

DSPFirst demo: hear a 5-point moving average

$$y[n] = (x[n] + x[n-1] + x[n-2] + x[n-3] + x[n-4]) / 5$$

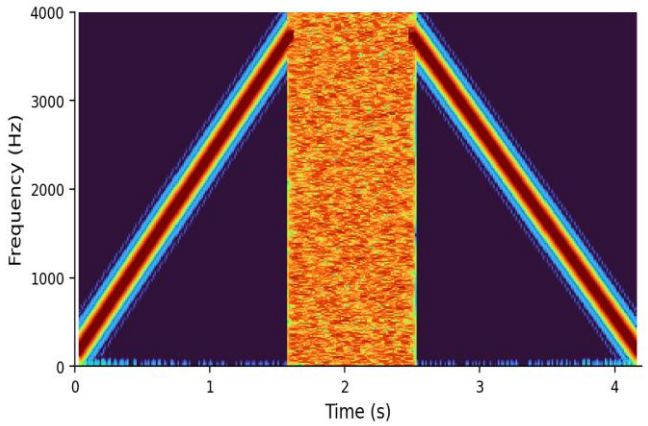
Magnitude response



Zeros: $\hat{\omega} = 2\pi/5$ and $4\pi/5$

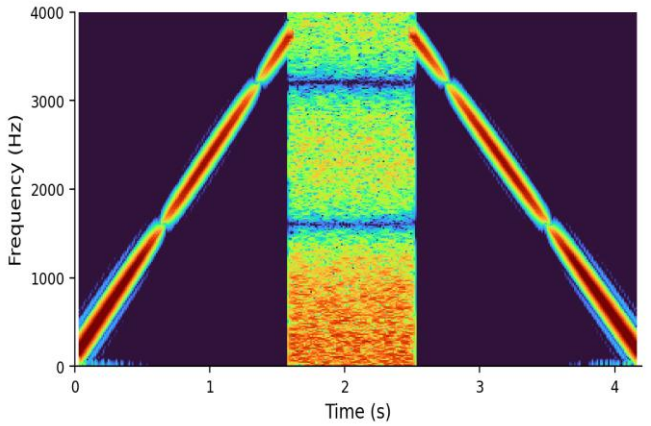
At f_s = 8 kHz: 1.6 and 3.2 kHz

Before filtering



Listen: input ↗

After filtering



Listen: output ↗

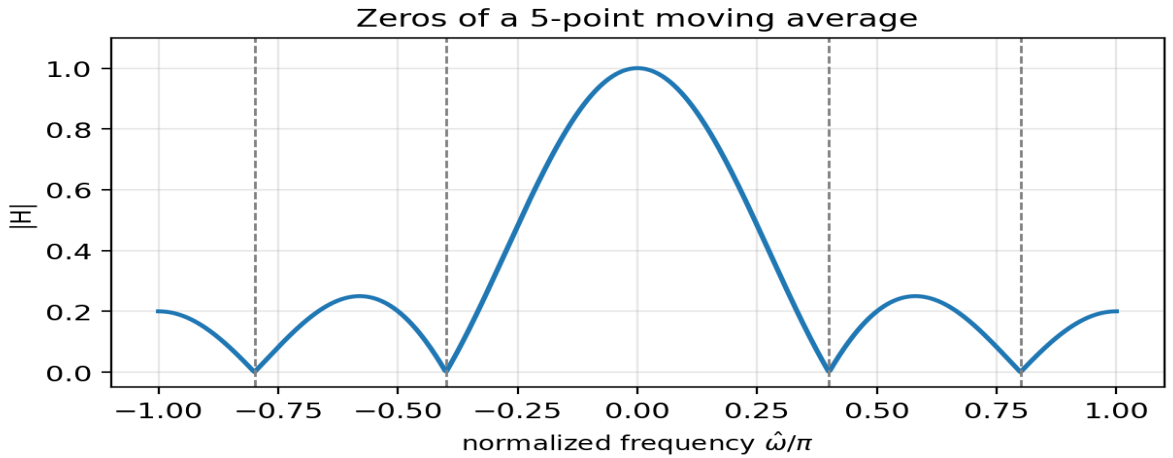
Open the full DSPFirst audio demo ↗

Example 6: moving-average zeros

Cancel a sinusoidal disturbance

Use an L-point moving average to cancel $\hat{\omega} = \pi/2$.

1. Zeros satisfy $\hat{\omega} = 2\pi m/L$, with $m=1,\ldots,L-1$.
2. Set $\pi/2 = 2\pi m/L \Rightarrow L = 4m$.
3. Smallest choice: $m=1$, so $L=4$.
4. Difference equation: $y[n]=(1/4)(x[n]+x[n-1]+x[n-2]+x[n-3])$.



Even- and odd-length moving averages

The symmetry centre of the averaging window determines the group delay

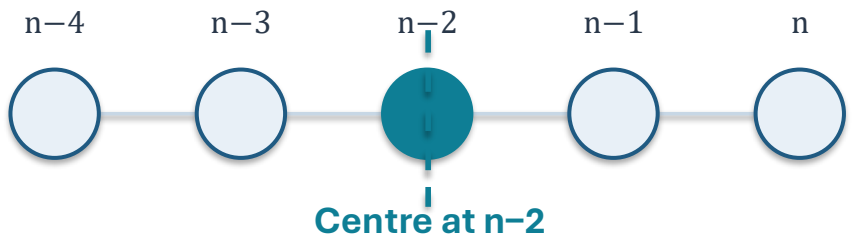
CAUSAL L-POINT MOVING AVERAGE

Equal weights on x[n], x[n−1], …, x[n−L+1]

Phase factor: $e^{-j\omega D}$ $\quad D = (L-1)/2$

ODD LENGTH · L = 5

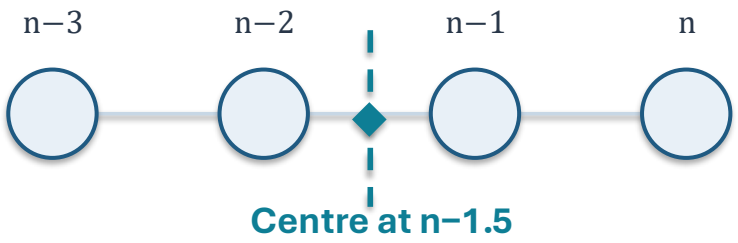
5 samples, each weighted 1/5



$D = 2$ samples

EVEN LENGTH · L = 4

4 samples, each weighted 1/4



$D = 1.5$ samples

Half-sample delay = a frequency-dependent phase shift; output samples are still computed at integer n.

Key idea. Odd L → integer delay; even L → half-integer delay. Both have linear phase.

Notation reference

Symbol	Meaning
$x[n]$, $y[n]$	Input and output discrete-time sequences
$h[n]$	Impulse response of the FIR filter
b_k	FIR filter coefficient at delay k
M	Filter order; the largest delay index
$L=M+1$	Filter length or number of coefficients
$H(e^{j\hat{\omega}})$	Frequency response evaluated at normalized radian frequency $\hat{\omega}$
$\lvert H(e^{j\hat{\omega}})\rvert$	Magnitude response or frequency-dependent gain
$\angle H(e^{j\hat{\omega}})$	Phase response
$\hat{\omega}$	Normalized radian frequency, in radians per sample
ω	Continuous-time radian frequency, in radians per second
T_s	Sampling period
$f_s=1/T_s$	Sampling rate
$*$	Convolution operator
$\delta[n]$	Unit impulse sequence
$u[n]$	Unit-step sequence

Key takeaways: FIR frequency response

1 One frequency in → the same frequency out

For an LTI filter, a sinusoid changes amplitude and phase, not frequency.

2 Read the frequency response

$|H(e^{j\hat{\omega}})|$ tells the gain; $\angle H(e^{j\hat{\omega}})$ tells the phase shift.

3 Predict the output of a sinusoid

Find $\hat{\omega}$, evaluate H at that frequency, then scale the amplitude and add the phase.

4 Understand the basic FIR examples

A delay changes phase; a first difference suppresses DC; a moving average smooths.

5 Combine filters and components

Filter each sinusoid separately (superposition); cascaded responses multiply.

6 Recognize zeros, transients, and sampling

$H = 0$ cancels a component; onset produces transients; $\hat{\omega} = 2\pi f/f_s$ connects Hz to digital frequency.

Practical rule: $y[n] = A|H(e^{j\hat{\omega}})| \cos(\hat{\omega}n + \phi + \angle H(e^{j\hat{\omega}}))$